

Name: Shaikh Mohammed Saad

Roll no: B-39

Branch:- AI&DS(B)

Experiment-3 : Data Ingestion and Cleaning

Aim:- Data Wrangling with Pandas: Manipulate and transform data to suit the analysis need.

Q1. What is data wrangling?. Explain the difference between Data Wrangling and Data preprocessing.

ANS:- Data wrangling is the process of cleaning, transforming, and organizing raw data into a more useful and structured format for analysis. It involves tasks such as handling missing values, filtering data, and reshaping datasets. Data preprocessing, on the other hand, is a broader term that includes all steps taken before applying machine learning algorithms, including data wrangling, feature scaling, encoding categorical data, and splitting datasets. In simple terms, data wrangling mainly focuses on cleaning and transforming data, while data preprocessing includes additional steps required to prepare data specifically for modeling.

```
import pandas as pd
data = pd.DataFrame({
    'Name': ['Ali', 'Sara', None, 'John'],
    'Age': [20, None, 22, 21],
    'Marks': [85, 90, None, 88]
})
print("Original Data:")
print(data)
data['Age'].fillna(data['Age'].mean(), inplace=True)
data['Marks'].fillna(0, inplace=True)
data.dropna(subset=['Name'], inplace=True)
print("\nAfter Data Wrangling:")
print(data)
from sklearn.preprocessing import MinMaxScaler
scaler = MinMaxScaler()
data[['Marks']] = scaler.fit_transform(data[['Marks']])
print("\nAfter Data Preprocessing (Scaling):")
print(data)
```

Original Data:

	Name	Age	Marks
0	Ali	20.0	85.0
1	Sara	NaN	90.0
2	None	22.0	NaN
3	John	21.0	88.0

After Data Wrangling:

	Name	Age	Marks
0	Ali	20.0	85.0
1	Sara	21.0	90.0
3	John	21.0	88.0

/tmp/ipykernel_1395/1020328632.py:9: FutureWarning: A value is trying to be set on a copy of a DataFrame or The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=Tr

```
data['Age'].fillna(data['Age'].mean(), inplace=True)
```

/tmp/ipykernel_1395/1020328632.py:10: FutureWarning: A value is trying to be set on a copy of a DataFrame or The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=Tr

```
data['Marks'].fillna(0, inplace=True)
```

After Data Preprocessing (Scaling):

	Name	Age	Marks
0	Ali	20.0	0.0
1	Sara	21.0	1.0
3	John	21.0	0.6

Q2. What is the importance of exploratory data analysis(EDA) in pandas?

ANS:- Exploratory Data Analysis (EDA) is the important step in Data analysis that help in understanding the structure, pattern, and relationship with dataset. In Pandas, EDA allow user to summarize the data using the statistical measure, visualize trends, and detect anomalies and outliers. It help in identifying missing value, understanding distribution, and making inform decisions about further data processing. EDA is essential because it provide insights into the data before applying any complex analysis or machine learning models.

```
import pandas as pd
data = pd.DataFrame({
    'Name': ['Ali', 'Sara', 'John', 'Aman'],
    'Age': [20, 21, 22, 23],
    'Marks': [85, 90, 88, 92]
})
print(data.info())
print(data.describe())
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 4 entries, 0 to 3
Data columns (total 3 columns):
#   Column  Non-Null Count  Dtype
---  -
0    Name         4 non-null     object
1    Age          4 non-null     int64
2    Marks        4 non-null     int64
dtypes: int64(2), object(1)
memory usage: 228.0+ bytes
None
```

	Age	Marks
count	4.000000	4.000000
mean	21.500000	88.750000
std	1.290994	2.986079
min	20.000000	85.000000
25%	20.750000	87.250000
50%	21.500000	89.000000
75%	22.250000	90.500000
max	23.000000	92.000000

Q3. What are the main differences between series, Dataframe, and Panel?

ANS:-

Feature	Series	DataFrame
Dimensionality	1-dimensional (1D)	2-dimensional (2D)
Data Structure	Homogeneous array with a labeled index	Heterogeneous tabular data with labeled axes
Analogy	A single column of a spreadsheet	An entire spreadsheet or SQL table
Contents	Stores a single data type	Can store different data types per column
Index/Axes	One axis (index)	Two axes (index and columns)
Use Case	Storing a single feature or time series data	Most common for general tabular data
Creation	<code>pd.Series()</code>	<code>pd.DataFrame()</code>

```
import pandas as pd
# Series (1D)
s = pd.Series([10, 20, 30])
# DataFrame (2D)
df = pd.DataFrame({
    'A': [1, 2],
    'B': [3, 4]
})
print(s)
print(df)
```

```
0    10
1    20
2    30
```

```
dtype: int64
   A B
0  1  3
1  2  4
```

Q4.Explain the use of:

a) isnull(): The isnull() function is used to check the missing or null value in a dataset. It returns a boolean result where the True indicates a missing value and False indicates a non-missing value.

```
import pandas as pd
df = pd.DataFrame({'A' : [1, None, 3]})
print(df.isnull())
```

```
   A
0  False
1   True
2  False
```

b) notnull(): The notnull() function is used to check for non-missing values in a dataset. It returns the True for values that are present and False for missing values.

```
import pandas as pd
df = pd.DataFrame({'A' : [1, None, 3]})
print(df.notnull())
```

```
   A
0   True
1  False
2   True
```

c) dropna(): The dropna() function is used to remove rows or columns that contain missing values. By default, it removes rows with at least one null value.

```
import pandas as pd
df = pd.DataFrame({'A' : [1, None, 3]})
print(df.dropna())
```

```
   A
0  1.0
2  3.0
```

d) fillna(): The fillna() function is used to replace the missing values with a specified value such as zero, mean, or any constant.

```
import pandas as pd
df = pd.DataFrame({'A' : [1, None, 3]})
print(df.fillna(0))
```

```
   A
0  1.0
1  0.0
2  3.0
```

Q5. what is column transformation and why is it important in data analysis?

ANS:- Column transformation refers to modifying or converting the values of the column to make them suitable for analysis. This may include data types, scaling values, applying mathematical operations, or creating new derived columns. Column transformation is important because it improves the quality and usability of the data, making it easier to analyse and interpret. It also plays a key role in preparing data for machine learning models, where properly formatted data leads to better performance and accurate results.

```
import pandas as pd
df = pd.DataFrame({'A': [1, None, 3]})
print(df.isnull())
print(df.notnull())
```

```
print(df.dropna())  
print(df.fillna(0))
```

```
      A  
0  False  
1   True  
2  False  
      A  
0   True  
1  False  
2   True  
      A  
0  1.0  
2  3.0  
      A  
0  1.0  
1  0.0  
2  3.0
```

Conclusion:- Data wrangling using Pandas is a crucial step in data analysis that helps convert raw data into a clean and structured format. Techniques like EDA, handling missing values, and column transformation improve data quality and make the analysis more accurate and reliable.